

Supervised Machine Learning Techniques for Predicting Titanic Survival

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Abstract

Supervised machine learning offers multiple approaches to binary classification, each with distinct trade-offs in accuracy, interpretability, and computational cost. This study compares three widely used classifiers, k-Nearest Neighbors (k-NN), Decision Trees, and Random Forests, on the Titanic survival prediction task. k-NN is a non-parametric, instance-based method that predicts class labels through majority voting among nearby points in feature space; it is conceptually simple but can be computationally expensive at inference time. Decision Trees learn hierarchical decision rules through recursive partitioning, providing transparent decision logic but often exhibiting high variance and a tendency to overfit. Random Forests reduce this variance via ensembling, aggregating many decorrelated trees to improve generalization.

Models were evaluated on an augmented Titanic dataset containing 891 passenger records and 26 base features, using a stratified 75/25 train-test split and cross-validated hyperparameter tuning. On the held-out test set, k-NN achieved the highest accuracy (80.72%) and an ROC-AUC of 0.8208, while the strongest Random Forest variant achieved 80.27% accuracy. Within the Random Forest family, a feature-selection pipeline achieved near-top accuracy (79.82%) with a dramatically smaller feature set (14 features), highlighting an accuracy–complexity trade-off. Overall, results show that ensemble methods provide strong performance and robustness, while carefully preprocessed distance-based methods can remain

highly competitive; simpler models retain value when interpretability or compactness is prioritized.

Introduction

Supervised machine learning is central to modern classification, where the objective is to assign discrete labels to unseen observations using patterns learned from labeled data. As real-world datasets have grown in size and complexity across domains such as healthcare, finance, natural language processing, and computer vision, hand-crafted rule systems have become difficult to scale and maintain. Supervised learning provides a data-driven alternative that can capture relationships between features and outcomes with measurable generalization performance.

In supervised classification, models learn a mapping from predictors to class labels using historical examples with known outcomes. After training, the learned mapping is applied to new observations to support decision-making under uncertainty. Because performance on unseen data is the primary goal, evaluation procedures such as stratified splitting and cross-validation are essential for estimating generalization and preventing optimistic bias.

Classification methods also differ in the trade-offs they impose. Some prioritize interpretability and transparent decision rules, which can be critical in high-stakes settings; others prioritize predictive power in complex, high-dimensional feature spaces. This study examines representative methods across this spectrum (k-NN, Decision Trees, and Random Forests) and compares their empirical performance on Titanic survival prediction.

Methodology

2.1 *k*-Nearest Neighbors (*k*-NN)

The workflow begins by loading *titanic_augmented.csv* into a pandas DataFrame, separating *Survived* as the target from all predictors, and performing a stratified 75/25 train-test split. Stratification preserves the survival class proportion across splits, improving the fairness and stability of evaluation.

Preprocessing is implemented in a reusable routine applied within each cross-validation fold and on the final train-test split to avoid leakage. High-cardinality or noisy identifier-like columns (e.g., *PassengerId*, *Name*, *Ticket*, *Cabin*, and engineered identifiers) are removed to reduce sparsity and limit dimensional growth. Missing values are imputed using training statistics only: numerical features are filled with the median, and categorical features are filled with the mode (or “Unknown” when a mode is undefined). Categorical variables are then one-hot encoded (*pd.get_dummies*). Because train and test sets may contain different category levels, the test matrix is reindexed to match the training columns, with unseen categories represented as zeros.

All features are standardized with *StandardScaler* to ensure comparable scales, which is essential for distance-based methods. Principal Component Analysis is then applied with *n_components = 0.95*, retaining 95% of the variance to reduce dimensionality introduced by one-hot encoding and improve *k*-NN efficiency. Hyperparameters are selected using stratified 5-fold cross-validation over the number of neighbors, distance weighting (uniform vs. distance), and distance metric (Euclidean vs. Manhattan). The final model is trained on the full training set using the best configuration and evaluated on the held-out test set using accuracy, precision, recall, F1, and ROC-AUC.

2.2 Random Forest

A consistent experimental design is used to support fair comparisons across Random Forest variants. The same stratified 75/25 train-test split is reused across all variants (*random_state* = 42), and hyperparameters are tuned only on the training portion using stratified 5-fold cross-validation (*StratifiedKfold(..., shuffle=True, random_state=42)*).

All preprocessing is implemented with scikit-learn *Pipeline* and *ColumnTransformer* to prevent leakage during cross-validation. Numeric features are median-imputed, and selected right-skewed variables are transformed with *log1p* after imputation to stabilize long-tailed distributions (notably fare-related features). Categorical features are imputed as a dedicated "Missing" category and one-hot encoded with *handle_unknown="ignore"* to prevent failures from unseen test categories. Model complexity is quantified by the number of transformed features after preprocessing (and after selection when applicable).

Three Random Forest strategies are compared. The **FeatureEngineering** variant adds structured predictors such as title extraction, family structure, cabin/deck indicators, ticket-group size, age/fare bins, and a *Sex_Pclass* interaction, while dropping high-cardinality and identifier-like fields to control dimensionality. The **Large** variant removes only clear identifiers (e.g., *PassengerId*, *service_id*, *booking_reference*) and retains most raw fields, producing a much larger one-hot feature space. The **CorrRFSelect** variant reduces dimensionality in two stages: numeric correlation screening on the training set followed by *SelectFromModel* using Random Forest importances (median threshold) after one-hot encoding. All variants are tuned with the same *GridSearchCV* grid (estimators, depth, split/leaf thresholds, feature subsampling, and class weighting) and evaluated on the held-out test set using accuracy and class-specific metrics.

2.3 Decision Tree

The augmented dataset undergoes domain-informed feature engineering to improve signal extraction while controlling sparsity. Engineered predictors include grouped title categories, a solo-passenger indicator, age discretization into life-stage bins, fare groupings, cabin deck extraction, and explicit missingness indicators for key fields (e.g., Age, Cabin, Embarked). High-cardinality identifiers (e.g., *Name*, *Ticket*, *Cabin*) and ID-like columns are excluded, and raw continuous variables are replaced with engineered categorical equivalents where appropriate.

The preprocessing pipeline applies median imputation to numeric features, logarithmic transformation to selected right-skewed variables (e.g., fare- and grouping-derived features), mode imputation to categorical features, and one-hot encoding. Data is split using a stratified 75/25 train-test partition to preserve class balance. Hyperparameters are optimized with *GridSearchCV* using stratified 5-fold cross-validation across depth, splitting constraints, feature subsampling, class weighting, impurity criterion, and cost-complexity pruning (*ccp_alpha*). Final evaluation on the held-out test set reports accuracy, confusion matrix, precision, recall, F1-score, and feature importance.

Experiments and Results

3.1 k-NN

The k-NN classifier with PCA-based dimensionality reduction achieves strong predictive performance on the Titanic survival task (Table 1). On the held-out test set, accuracy reaches 80.72% with ROC-AUC of 0.8208, indicating good discriminative capacity. However, recall is

lower (72.09%), implying a non-trivial false-negative rate for survivors. These results are consistent with known properties of k-NN: performance depends heavily on feature scaling and the geometry of the representation, while inference can be costly due to distance computations (Cover & Hart, 1967; Indyk & Motwani, 1998; Pestov, 2012). PCA helps mitigate the curse of dimensionality introduced by one-hot encoding by compressing the feature space while preserving variance, which can improve both stability and efficiency (Zhao et al., 2022).

Metric	Value
KNN Results (Test Set)	
accuracy	0.8072
precision	0.7654
recall	0.7209
f1	0.7425
roc_auc	0.8208

Table 1. The dataset has 891 rows and 26 columns. It was split into a training set of 668 rows and a test set of 223 rows. 27 PCA features were used for the analysis.

3.2 Random Forest

Random Forest results reveal a clear accuracy–complexity trade-off across variants. The FeatureEngineering model achieves CV accuracy of 0.8338 but drops to 0.7803 on the test set with 75 features. The Large model attains the best Random Forest test accuracy (0.8027) but

expands to 1,378 features due to extensive one-hot encoding. CorrRFSelect remains highly competitive (0.7982 test accuracy) while reducing the final feature set to 14, indicating that substantial dimensional compression is possible with only a minor loss in accuracy. The Large model's advantage likely stems from retaining high-cardinality raw fields that contain weak but exploitable signals; however, that gain comes with increased complexity and reduced interpretability. The FeatureEngineering drop from CV to test suggests that discretizations and interactions did not generalize cleanly and may overlap with existing augmented signals. CorrRFSelect performs well because explicit pruning limits one-hot noise and concentrates the model on stronger predictors.

3.3 Decision Tree

The tuned Decision Tree achieves 77.58% test accuracy, with cross-validation performance of 82.04%, indicating overfitting despite depth and leaf-size constraints. Class-wise metrics show stronger performance on non-survivors than survivors, consistent with the high-variance behavior of single trees and sensitivity to class structure (Hastie et al., 2009). Feature importance emphasizes title-based predictors and socioeconomic proxies, aligning with historical survival dynamics, while the observed generalization gap supports the standard motivation for ensemble methods such as Random Forests (Breiman, 2001).

4. Conclusions

4.1 k-NN

k-NN achieves the strongest overall test accuracy (80.72%) and ROC-AUC (0.8208), demonstrating that careful scaling and PCA-based dimensionality reduction can make

distance-based methods competitive on augmented, mixed-type tabular data. The relatively lower recall (72.09%) indicates under-identification of survivors, which may be undesirable in settings where false negatives are costly. k-NN remains a strong baseline, though sensitivity to representation should be considered.

4.2 Random Forest

Random Forest variants demonstrate a pronounced performance–complexity trade-off. The Large model delivers the highest Random Forest test accuracy (80.27%) but requires a 1,378-feature representation, while CorrRFSelect achieves nearly identical performance (79.82%) using only 14 features, yielding a substantial reduction in dimensionality with minimal accuracy loss. FeatureEngineering performs well in cross-validation (83.38%) but underperforms on the held-out test set (78.03%), suggesting weaker generalization. For an efficiency-oriented objective, CorrRFSelect is the most defensible Random Forest choice.

4.3 Decision Tree

The Decision Tree provides interpretable decision logic and identifies meaningful predictors, but exhibits a generalization gap consistent with high variance in single-tree models. Test accuracy (77.58%) is lower than both k-NN and Random Forest, and survivor recall remains relatively weaker. Decision Trees are valuable for interpretability and feature insight, but ensemble methods are better suited when generalization performance is prioritized.

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